FDTD-Based Optical Scatterometry for Trench Critical Dimension Metrology: Inverse Problem Optimization for Semiconductor Nanostructures

Project Overview

This project focused on the use of optical scatterometry, Ansys Lumerical FDTD simulations, and optimization algorithms to estimate the critical dimensions of trench nanostructures used in semiconductor-related applications.

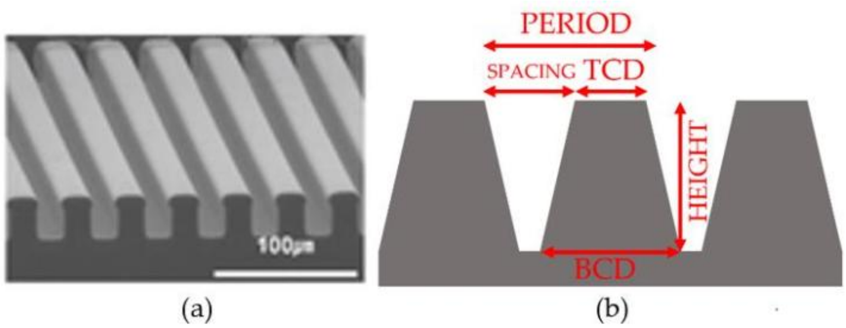

Figure 1. (a) SEM trench image. (b) Typical trench critical dimensions.

The main goal was to solve an inverse metrology problem: instead of directly measuring the nanostructure geometry, the optical response of the structure was simulated and compared with experimental data. The trench dimensions were then adjusted through optimization algorithms until the simulated response closely matched the measured result.

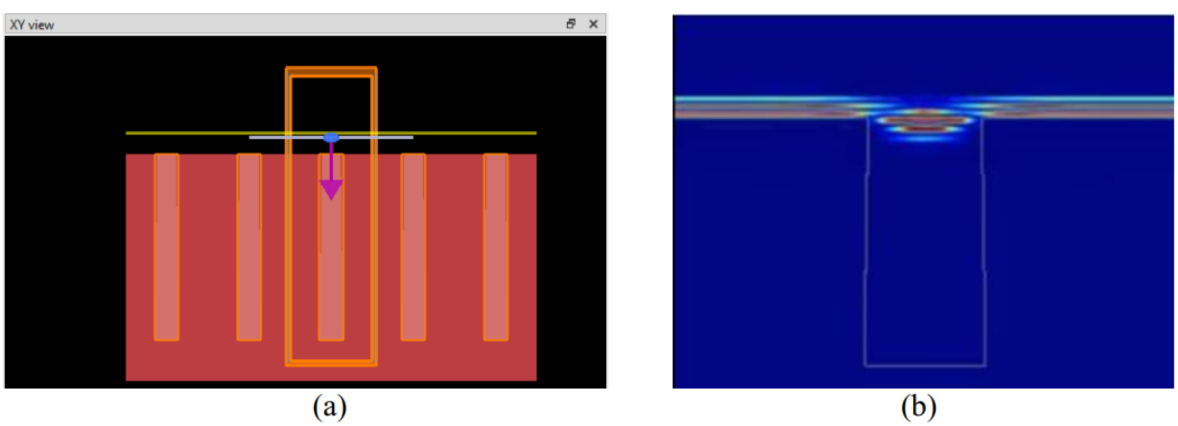

Figure 2. (a) FDTD trench simulation set-up. (b) Visual representation of the FDTD simulation.

The estimated parameters included top critical dimension, bottom critical dimension, spacing, height, and period.

Motivation

Accurate critical dimension metrology is essential in semiconductor manufacturing because small deviations in nanoscale structures can affect device performance, yield, and reliability. However, direct measurement techniques such as SEM can be time-consuming, sample-dependent, or limited in scalability.

Simulation-assisted optical metrology provides a non-destructive alternative. By combining experimental optical measurements with electromagnetic simulations, it is possible to estimate structural parameters without relying only on direct physical inspection.

This project explored how FDTD simulations and optimization algorithms can be combined to improve the estimation of trench geometries from optical measurement data.

Methodology

The workflow consisted of four main stages.

First, several metaheuristic optimization algorithms were studied and implemented. These included Genetic Algorithm, Particle Swarm Optimization, Tabu Search, Differential Evolution, Ant Colony Optimization, Simulated Quantum Annealing, and Simulated Annealing.

Second, trench structures were modeled using Ansys Lumerical FDTD. The simulations were used to calculate the optical response of different trench geometries.

Third, experimental and simulated data were preprocessed and normalized to make the comparison more reliable. A comparison metric was then used to evaluate how closely the simulated response matched the measured data.

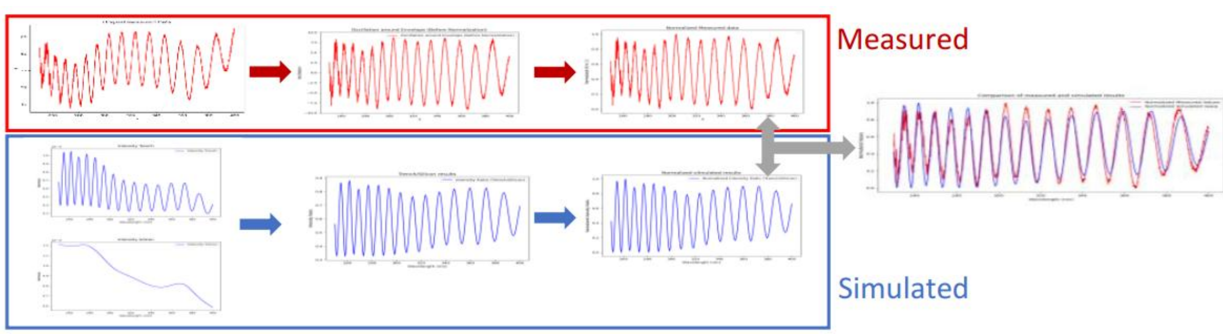

Figure 3. Experimental and simulated data preprocessing and final comparison.

Finally, the FDTD simulation, data comparison, and optimization algorithms were integrated into an automated Python workflow. The optimization algorithm adjusted the trench parameters and repeated the simulation process until the best match was found.

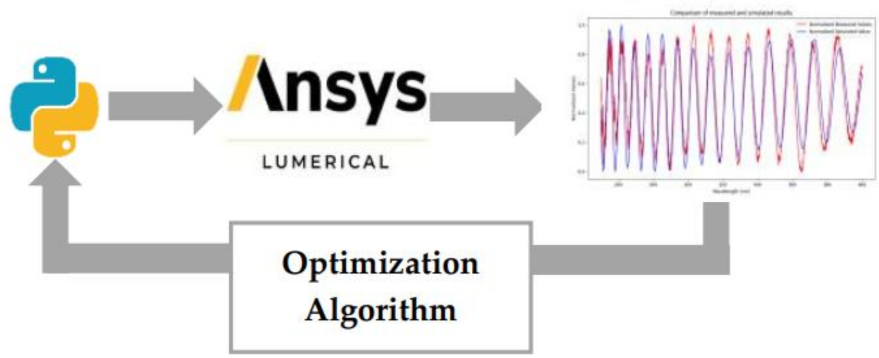

Figure 4. Automated optimization workflow linking Python, Ansys Lumerical, and the comparison metric.

Technical Workflow

The complete workflow can be summarized as:

1. Define initial parameter ranges for the trench geometry.
2. Generate a candidate geometry using an optimization algorithm.
3. Run the FDTD simulation in Ansys Lumerical.
4. Extract the simulated optical response.
5. Preprocess and normalize simulated and measured data.
6. Compare both signals using a similarity metric.
7. Update the candidate parameters through the optimization algorithm.
8. Estimate the critical dimensions that provide the best simulation-to-experiment match.

Results

The method was tested on two trench samples provided by an industrial partner. For each sample, seven optimization algorithms were executed and compared.

The results showed that all implemented algorithms were able to converge toward similar solutions. The estimated critical dimensions were compared with SEM reference values, showing that the simulation-assisted approach could recover the trench parameters with good agreement.

A key observation was that the main difference between the algorithms was not the final accuracy, but the execution time required to reach a comparable solution. Some algorithms required longer simulation times, while others produced similar results more efficiently.

For both trench samples, the estimated critical dimensions showed good agreement with SEM reference values, with most parameter differences remaining within approximately 0–2% depending on the optimization algorithm and sample geometry. These results demonstrate the potential of combining FDTD simulations and optimization algorithms for non-destructive optical critical dimension metrology.

My Role

My contributions to this project included:

- Implemented and compared seven metaheuristic optimization algorithms.
- Learned and configured Ansys Lumerical FDTD simulations for trench nanostructures.
- Processed and normalized experimental and simulated optical measurement data.
- Developed Python scripts to automate the simulation, optimization, and comparison workflow.
- Evaluated the estimated critical dimensions against SEM reference values.
- Compared the performance of different optimization algorithms in terms of accuracy and execution time.
- Prepared the final technical report and presentation summarizing the methodology and results.

Key Technical Areas

- Optical scatterometry
- Optical critical dimension metrology
- FDTD simulation
- Ansys Lumerical
- Python automation
- Inverse problems
- Simulation-to-experiment matching
- Semiconductor nanostructure metrology
- Optimization algorithms
- Parameter estimation

Project Status

This project was completed during my first semester at the NTU Precision Metrology Laboratory as part of ongoing research on optical critical dimension metrology. The developed workflow demonstrated the feasibility of using FDTD-based simulation and optimization algorithms for trench critical dimension estimation.

Confidentiality Note

This project is presented as an academic and technical portfolio summary. Some raw data, implementation details, and experimental files are simplified or omitted due to confidentiality.